Assignment: End-to-End Data Preprocessing & Feature Engineering on Google Play Store Dataset

Dataset

Google Play Store Apps Dataset

Problem Statement

You are working as a Data Scientist at Google Play Store Analytics Team.

The raw dataset contains missing values, inconsistent formats, outliers, categorical variables, and noisy data. Your task is to perform complete Data Preprocessing and Feature Engineering before the dataset can be used for Machine Learning.

Learning Outcomes

After completing this assignment, students should be able to:

- Understand dataset structure
- Handle data quality issues
- Handle missing values
- Handle duplicate records
- Handle outliers
- Perform feature transformation
- Encode categorical variables
- Scale numerical features
- Extract features from dates and text
- Create new features
- Select important features
- Prepare final dataset for Machine Learning

Phase 1: Data Collection & Understanding

Tasks

1. Load the dataset.
2. Display the first 10 records.
3. Display the last 10 records.
4. Check the shape of the dataset.
5. Display all column names.
6. Check data types of all columns.
7. Generate summary statistics.
8. Identify:

    Numerical Features
    Categorical Features

Questions

1. How many rows and columns are present?
2. Which feature has the highest number of unique values?
3. Which column may act as an identifier?
4. Which columns appear to need cleaning?

Phase 2: Data Quality Assessment

Tasks

1. Check for duplicate records.
2. Count duplicate rows.
3. Remove duplicate rows.

4. Verify the dataset shape after removing duplicates.

Questions

1. How many duplicate records were found?
2. Why should duplicate records be removed before model building?
3. What problems can duplicate data create?

Phase 3: Data Type Correction

The following columns are stored as text:

```
Installs
Price
Size
Reviews
```

Tasks

Convert values such as:

```
1,000+
10,000+
1M
20M
Varies with device
$4.99
```

into meaningful numerical values.

Questions

1. Why are these columns unsuitable for Machine Learning in their current form?
2. What preprocessing steps were required to convert them?
3. Which column was the most challenging to clean and why?

Phase 4: Missing Value Treatment

Tasks

1. Identify missing values.
2. Calculate the percentage of missing values in each column.
3. Categorize columns based on missing value percentage.

Numerical Columns

Apply:

- Mean Imputation
- Median Imputation

Compare the results.

Categorical Columns

Apply:

- Mode Imputation
- Constant Value Imputation

Questions

1. Which columns contain the highest number of missing values? 2.
Which imputation method is most suitable for the Rating column? 3.
Why is handling missing data important?

Phase 5: Handling Inconsistent Data

Tasks

Identify and clean:

- Currency symbols
- Commas
- Plus signs

- Text values inside numerical columns
- Invalid entries
- Special characters

Examples:

```
1,000+
$4.99
19M
Varies with device
```

Questions

1. Why is data consistency important?
2. What issues arise when numerical columns contain text values?
3. How did you handle inconsistent records?

Phase 6: Outlier Detection & Treatment

Features

Analyze:

```
Rating
Reviews
Installs
Price
```

Tasks

1. Visualize outliers using boxplots.
2. Detect outliers using the IQR method.
3. Detect outliers using Z-score.
4. Remove outliers.
5. Apply outlier capping.
6. Compare both approaches.

Questions

1. Which feature contains the most outliers?
2. Which outlier treatment method performed better?
3. Why can outliers negatively affect model performance?

Phase 7: Feature Transformation

Tasks

Check skewness of:

```
Reviews
Installs
Price
```

Apply:

Log Transformation

```
np.log1p()
```

Square Root Transformation

```
np.sqrt()
```

Questions

1. Which features were highly skewed?
2. Which transformation reduced skewness most effectively? 3.
Why is skewness a problem for some machine learning models?

Phase 8: Feature Encoding

Tasks

Apply Label Encoding on:

```
Type
```

Apply One-Hot Encoding on:

```
Category
Content Rating
Genres
```

Questions

1. Why is One-Hot Encoding preferred for Category and Genres?
2. When should Label Encoding be used?
3. What are the disadvantages of One-Hot Encoding?

Phase 9: Feature Scaling

Tasks

Apply:

StandardScaler

MinMaxScaler

on:

```
Rating
Reviews
Installs
Price
```

Questions

1. Compare the outputs of both scalers.
2. Which algorithms benefit from scaling?
3. Which scaler would you choose for this dataset and why?

Phase 10: Feature Extraction

A. Date Feature Extraction

Column:

```
Last Updated
```

Convert it into datetime format and extract:

- Year
- Month
- Day
- Quarter
- Day Name

Questions

1. Which year contains the highest number of app updates?
2. Which month contains the highest number of updates?

B. Android Version Extraction

Extract the minimum Android version from values such as:

```
4.1 and up
5.0 and up
6.0 and up
```

Create:

```
Minimum_Android_Version
```

Questions

1. Which Android version appears most frequently?
2. Why might this feature be useful?

Phase 11: Feature Creation

Create the following features.

Feature 1

```
Reviews_Per_Install
```

Formula:

```
Reviews / Installs
```

Feature 2

```
Paid_App_Flag
```

```
Paid = 1
Free = 0
```

Feature 3

```
App_Age
```

Formula:

```
Current Year - Update Year
```

Feature 4

```
Popularity_Score
```

Formula:

```
Rating * Reviews
```

Feature 5

```
Revenue_Potential
```

Formula:

```
Price * Installs
```

Feature 6

```
Category_Average_Rating
```

Calculate the average rating for each category and assign it to every app.

Questions

1. Which category has the highest average rating?
2. Which category has the highest revenue potential?
3. Are paid apps more popular than free apps?
4. Which newly created feature seems most useful?

Phase 12: Feature Selection

Assume:

```
Rating
```

is the target variable.

Tasks

1. Perform Correlation Analysis.
2. Identify highly correlated features.
3. Apply SelectKBest.
4. Apply Random Forest Feature Importance.
5. Compare results from all methods.

Questions

1. Which features have the strongest relationship with Rating?
2. Which features can be removed?
3. Why is feature selection important?

Phase 13: Train-Test Split

Tasks

1. Separate features (X) and target variable (y).
2. Use:

```
Rating
```

as the target.
3. Perform an 80:20 Train-Test Split.
4. Display the shapes of:

    X_train
    X_test
    y_train
    y_test

Questions

1. Why do we split data before training?
2. What problems occur if we train and test on the same dataset?

Phase 14: Build a Preprocessing Pipeline

Using:

```
ColumnTransformer
Pipeline
```

Create an automated preprocessing pipeline that performs:

    Missing Value Handling
    Encoding
    Scaling

in a single workflow.

Questions

1. What are the advantages of using a Pipeline?
2. How does a Pipeline help prevent data leakage?

Final Deliverables

Submit:

1. `google_playstore_cleaned.csv`
2. `google_playstore_feature_engineered.csv`
3. Jupyter Notebook (`.ipynb`)
4. A report containing:

    Challenges faced during preprocessing
    Important features discovered
    Key business insights
    Final conclusions

Reflection Questions

1. Which preprocessing step had the greatest impact on the dataset?
2. Which feature engineering technique was most useful?
3. Which newly created feature would you use in a machine learning model and why?
4. What challenges did you face while working with real-world data?
5. If you were building an App Rating Prediction System, which features would you select and why?